A new hollowing process for rapid prototype models

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Keywords

Boolean functions, Rapid prototypes, Algorithmic languages

Abstract

A new method of hollowing rapid prototype models based on STL models and their cross-sectional contours is presented to meet the demands of hollowed prototypes in casting and rapid prototype manufacturing. Offsetting along the *Z*-axis and cross-sectional contour offsetting are employed to perform the hollowing operation. The process performs two-dimensional Boolean operations on the polygons made by the offset contours of cross-sectional contours instead of three-dimensional offsetting of the STL models. This hollowing operation is especially suitable for hollowing STL models with free-form surfaces. Detailed algorithms are described to generate the correct offset contours of an STL model. Adopting this method, the hollowing process is dramatically simplified and becomes more efficient. This method has been verified by practical case studies, and it is proved that this simplified hollowing operation can reduce the prototype build time and cost.

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1. Introduction

Rapid prototyping (RP) can directly convert CAD models into three-dimensional (3D) physical solid objects, faster than conventional manufacturing processes. RP techniques build the prototypes layer by layer. The tracing of the cross-sectional area of a part on each layer is the most time-consuming process (Li *et al.*, 1998). If a thin-shell prototype possesses enough mechanical strength, then a hollowing process can be used to reduce the solid volume to be built. Hence, material is saved and the building efficiency is enhanced greatly.

One of the main applications of rapid prototypes is to create master patterns for different casting processes, where thin-shell prototypes are demanded. For example, in SL prototypes, their accuracy lends itself well to investment casting to produce functional metal parts. In investment casting, thin-shell prototypes are needed, whose internal architecture is designed as described Hague *et al.* (2001). This allows them to collapse inwardly, so as to avoid fracturing of the ceramic shell.

Therefore, there is an important need to produce hollowed prototypes. Several different processes have been employed to produce hollowed objects for RP. They are:

(1) spatial enumeration methods,
(2) constructive solid geometry (CSG) and boundary representation (B-Rep) offsetting methods,
(3) STL model offsetting methods, and
(4) curve offsetting methods.

Chiu and Tan (1998) employ one-dimensional Boolean operations between the ray representations of the model and voxel elements to hollow an STL model, instead of performing 3D Boolean operations. The use of the enumeration method causes the internal staircase effect (Koc and Lee, 2002). The CSG technique is used to gain a thin-shell solid by subtracting the original solid from its offset counterpart (Lam *et al.*, 1997; Li *et al.*, 1998). Li *et al.* (1998) presented an algorithm based on CSG to implement the solid offsetting, which can be extended to other solid modeling schemes. Zuo *et al.* (1999) implemented a 2D merging operation of loops in a surface's parametric region to find the shell for a B-Rep solid. However, their methods cannot be used after converting the CAD model of a part to an STL model (Koc and Lee, 2002). The direct generation

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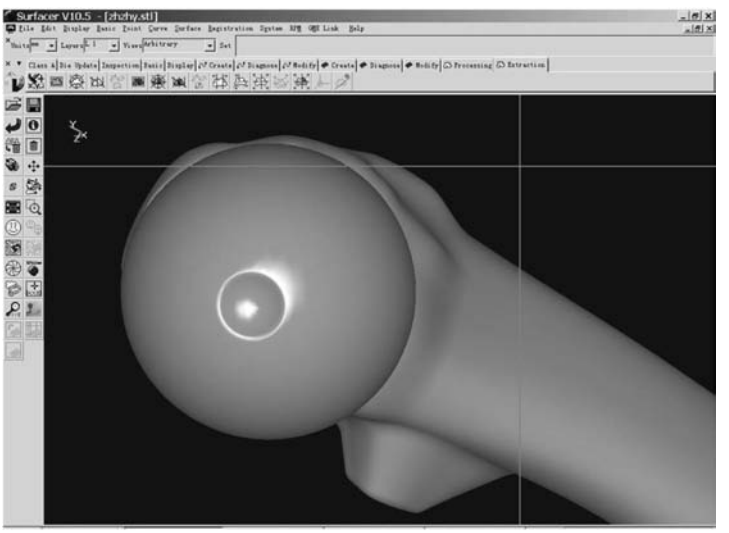
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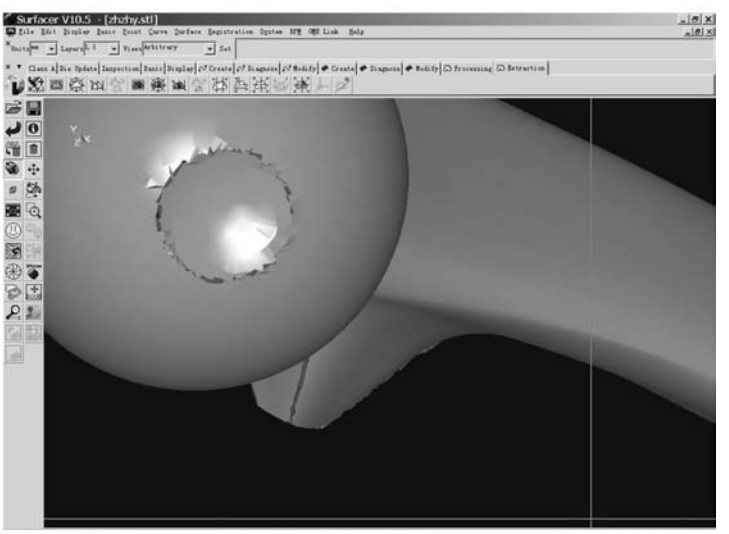
of STL models from scanned data has obvious advantages in that it can avoid the time penalty and errors often associated with the CAD modeling process (Horváth and Yang, 2002; Lee *et al.*, 2002; Zhang *et al.*, 2002). As a result, a hollowing process for STL models is highly desirable.

However, when an STL model (for example, Figure 1(a)) is negatively offset it can cause problems. In this case, when an offset distance of more than 0.8 mm was used in Surfacer®V10.5, the offset STL model contained gaps in sharp corners (Figure 1(b)), which cause trouble during the slicing process of an RP system. Yau *et al.* (2003) found that the rectification of such an STL model is not an easy task.

Figure 1 Offset operation in Surfacer®V10.5: (a) artificial proximal femur's STL model; and (b) artificial proximal femur's offset STL model with gaps



(a)



(b)

Ganesan and Fadel (1994) offset a sliced STL model to create a hollow part using a constant offset for each slice. This is not suitable for creating hollow parts that have varying surface normals and this method has a major drawback in that when offsetting the cross-sectional contours, the part's surfaces that are nearly parallel to the slicing planes are lost. Koc and Lee (2002) developed a non-uniform offsetting and hollowing method using an averaged surface normals method to find the correct offset contours for an STL model. However, when the different vertices of an STL model are offset along their averaged surface normal, some of them may be superposed on one another. Thus, the offset STL model may contain some triangular facets with overlaps and inconsistent orientations.

In order to overcome the aforementioned problems, a new hollowing process for rapid prototyping is introduced in this paper. In the following sections, details of this method will be described. Some definitions and basic conceptions are given in Section 2. Section 3 contains a formal description of the offsetting along the *Z*-axis and the cross-sectional contour offsetting, followed by the generation of the offset contours. Some practical cases are discussed in Section 4. Finally, conclusions are given in Section 5.

2. Preliminary definitions and basic conceptions

This section introduces some definitions and basic conceptions.

Definition 1. If the angle between the normal of a triangle facet and *Z*-axis is within the interval $[-1°, +1°]$, this triangle facet is called a horizontal facet *HF*.

Definition 2. Given a horizontal facet HF_i, its normal vector is (N_x^i, N_y^i, N_z^i), and when hollowing, HF_i moves toward the direction $(0, 0, -N_z^i)$ with a given hollowing thickness h. Then the translational movement along the *Z*-axis is defined as the action region ar_i of HF_i. If $N_z^i > 0$, then:

$$ar_i = [HFz_{\min} - h, HFz_{\max}] \tag{1}$$

also if $N_z^i < 0$, then:

$$ar_i = [HFz_{\min}, HFz_{\max} + h] \tag{2}$$

where $HFz_{\max}$ and $HFz_{\min}$ are the maximum and minimum *Z*-coordinate values of HF_i.

The STL model slicing process is a key part of RP and manufacturing. First, it uses a series of planes to cut the part's STL model, and then the cross-sectional contours are created. In this paper,

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the slicing process employs some basic ideas from Hu and Li's (2000) algorithms, but it is somewhat modified in that the normal vectors of the triangle facets are stored when forming the edges of cross-sectional contours.

On each layer, all the regions surrounded by the cross-sectional contours along their specified orientations are the part's cross-sectional area on this layer. The features of a cross-sectional contour are:

(1) *no self-intersection*: any two edges of the same cross-sectional contour will intersect only if they are adjacent, and in this case the intersection point is a common vertex.
(2) *closure*: each edge of the same cross-sectional contour must be determined by two adjacent vertices, and each vertex is the intersection point of the two edges.
(3) *orientation*: each edge of a cross-sectional contour must have a specified direction and the directions of all edges must remain clockwise or anti-clockwise in sequence.
(4) *no intersection*: all the cross-sectional contours on the same layer must not intersect one another.

These features will ensure that each cross-sectional contour makes a planar polygon. Given a layer l_i, the set $f = \{f_1, f_2, \ldots, f_n\}$ consists of anti-clockwise cross-sectional contours on l_i, whose elements constitute the polygon set $F = \{F_1, F_2, \ldots, F_n\}$; and the set $c = \{c_1, c_2, \ldots, c_k\}$ consists of clockwise cross-sectional contours on l_i, whose elements constitute the polygon set $C = \{C_1, C_2, \ldots, C_k\}$.

Definition 3. On a given layer l_i, the part region PR_i is all the regions surrounded by the cross-sectional contours along their specified orientations.

Definition 4. On a given layer l_i, the element F_j in set $S_i = \{F_j \in F | \forall d \neq j, F_j \not\subset F_d\}$ is called the local MAX region on l_i. The polygon set made by all the anti-clockwise contours in F_j is defined as $F^j = \{F^j_m \in F | F^j_{m+1} \subset F^j_m \subseteq F_j\}$.

不包含在其他轮廓里面的逆时针轮廓。

Definition 5. Given a local MAX region F_j, a region cell of F_j is defined as:

$$A^j_m = F^j_m - \bigcup_{k=1}^{n(j)} C^{j,m}_k, \tag{3}$$

介绍一个待打印区域的定义，由一个外轮廓和多个内轮廓围成。

where $C^{j,m}_k$ is an element of polygon set $C^{j,m} = \{C^{j,m}_k \in C | C^{j,m}_k \subset F^j_m \backslash F^j_{m+1}\}$ and $n(j)$ is the number of all elements in set $C^{j,m}$. The clockwise cross-sectional contour making polygon $C^{j,m}_k$ is called the internal contour of region cell A^j_m, denoted as $c^{j,m}_k$; and the anti-clockwise cross-sectional contour making F^j_m is called the external contour of A^j_m, denoted as f^j_m.

The earlier definitions allow the establishment of a relationship between the part region and the region cells on each layer. On a given layer l_i, let the number of all elements in set S_i be Num, and the number of all region cells in local MAX region F_j be q. Then the part region PR_i can be calculated as follows:

$$PR_i = \bigcup_{j=1}^{\mathrm{Num}} \bigcup_{m=1}^{q} A^j_m \tag{4}$$

Figure 2 shows an example of application of equation (4), in which the relationship between the part region PR_i and the region cells on a given layer l_i can be observed, and the arrowheads indicate the orientation of the cross-sectional contours. PR_i consists of all the shadowed areas; clockwise cross-sectional contours $c^{1,1}_1, c^{1,1}_2, c^{1,1}_3, c^{2,1}_1$ make polygons $C^{1,1}_1, C^{1,1}_2, C^{1,1}_3, C^{2,1}_1$, and anti-clockwise cross-sectional contours f^1_1, f^1_2, f^2_1 make polygons F^1_1, F^1_2, F^2_1. The region cells are:

$$A^1_1 = F^1_1 - \left(C^{1,1}_1 \bigcup C^{1,1}_2 \bigcup C^{1,1}_3\right)$$
$$A^1_2 = F^1_2$$
$$A^2_1 = F^2_1 - C^{2,1}_1$$

All the shadowed areas (that is, part region PR_i) are $A^1_1 \bigcup A^1_2 \bigcup A^2_1$.

Once equations (3) and (4) are established, the offset result for the part region on a given layer can be obtained using Boolean operations on the polygons made by the offset contours of all the cross-sectional contours on this layer. The algorithms presented by Rivero and Feito (2000) can be employed to perform the Boolean operations between the general planar polygons (that is, manifold and non-manifold, with or without holes).

Definition 6. Given an edge of a cross-sectional contour, the normal of the triangle facet on which this edge lies is called the edge vector. In particular, when an edge is shared by two triangle facets, their averaged normal is used as the edge vector.

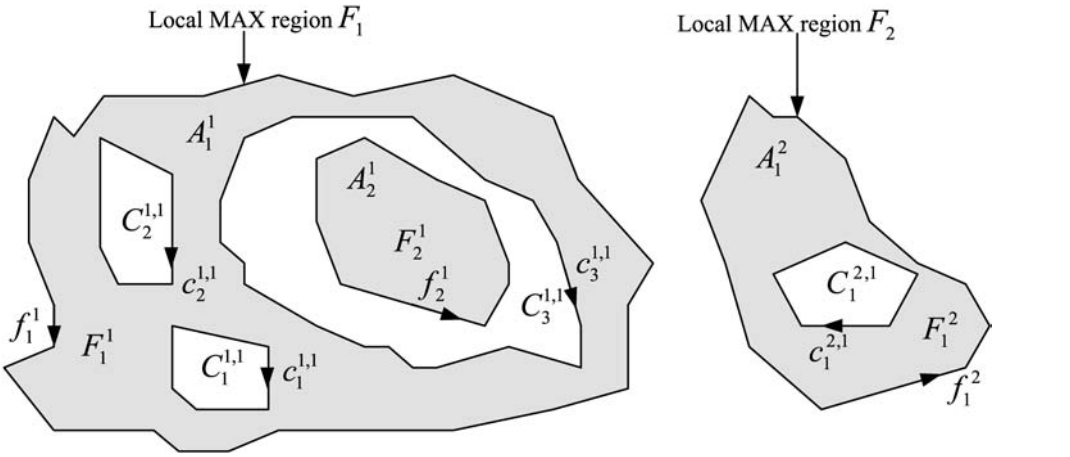
Definition 7. The offset edge vector is the projection of the inverse direction edge vector onto the XY plane.

3. Hollowing process for rapid prototype

Using a part's STL model and its cross-sectional contours, the hollowing operation for RP is divided into two steps: first, offsetting along the Z-axis is followed by cross-sectional contour offsetting. This operation can completely remove the difficulties of hollowing STL models with free-form surfaces.

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Figure 2 Representation of a part region through region cells



3.1 Offsetting along the *Z*-axis

When an STL model is undergoing the slicing process, the horizontal facets and their action regions can be determined according to the Definitions 1 and 2. The operations of the offsetting along the *Z*-axis are as follows.

(1) Finding all horizontal facets in the STL model and calculating their action regions.

(2) Performing an union operation on the action regions of all the horizontal facets, and let the result of this operation be denoted as AR^u and the number of all action regions be denoted as $n(ar)$, then:

$$AR^u = \bigcup_{i=1}^{n(ar)} ar_i. \tag{5}$$

(3) Generating the offset part regions on the layers inside AR^u after the offset part regions on the layers outside AR^u are obtained. The details are mentioned in Section 3.3.

In order to obtain the offset part regions, the cross-sectional contour offsetting operation is introduced.

3.2 Cross-sectional contour offsetting

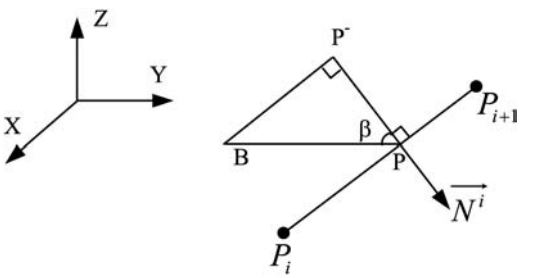
To offset all the internal contours and the external contours of each region cell in each local MAX region on a given layer and to obtain the offset part region on this layer, cross-sectional contour offsetting operation is performed.

3.2.1 Offset contours generation

The offset for cross-sectional contours is created in the horizontal plane, but the thickness of the hollowed prototype is along the inverse normal of its STL model.

As Figure 3 shows $\vec{N}^i$ is the vector of the edge $\overrightarrow{P_iP_{i+1}}$ of a cross-sectional contour; $\overline{|PP'|}$ indicates the specified hollowing wall thickness; $\overrightarrow{|PB|}$

Figure 3 Offset of a cross-sectional contour edge



indicates the wall thickness of the hollowed prototype projected on to the *XY* plane. Then the offset of $\overrightarrow{P_iP_{i+1}}$ is obtained by:

$$\overrightarrow{|PB|} = \frac{h}{\cos\beta} \tag{6}$$

where β is the angle between $\vec{N}^i$ and *XY* plane. If two triangular facets share the edge $\overrightarrow{P_iP_{i+1}}$, then the offset of $\overrightarrow{P_iP_{i+1}}$ can be estimated from:

$$\overrightarrow{|PB|} = \frac{h}{2(\cos\beta_1 + \cos\beta_2)} \tag{7}$$

where β_1 and β_2 are the angles between the normals of these two triangle facets and *XY* plane.

Other offset algorithms for 2D point-sequence curves are available (Choi and Park, 1999; Maekawa, 1999; Park and Chung, 2003). However, the offset distance for the edges of the cross-sectional contours calculated by equations (6) or (7), is not constant and so, a variable offsetting method has been presented.

In this variable offsetting, each edge in the cross-sectional contour is moved a given distance in the direction of its corresponding offset edge vector. Thus, the offset line of each edge can be obtained. The intersection points of the adjacent offset lines need to be calculated in turn, and these

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intersection points become the vertices of the offset contours.

As Figure 4 shows, P_{i-1}, p_i and P_{i+1} are the adjacent vertices of a contour on a given region cell A_m^j; $\overrightarrow{Q_1'P_i'}$ and $\overrightarrow{P_i'Q_2'}$ are the offset lines of the two edges $\overrightarrow{P_{i-1}P_i}$ and $\overrightarrow{P_iP_{i+1}}$. Let the vectors of $\overrightarrow{P_{i-1}P_i}$ and $\overrightarrow{P_iP_{i+1}}$ be $(N_x^{i-1}, N_y^{i-1}, N_z^{i-1})$ and (N_x^i, N_y^i, N_z^i), and then their offset vectors $\vec{u}_{i-1}$ and $\vec{u}_i$ are $-(N_x^{i-1}, -N_y^{i-1}, 0)$ and $-(N_x^i, -N_y^i), 0$, respectively. Then:

$$\overrightarrow{OQ_1'} = \overrightarrow{OP_i} + a\frac{\overrightarrow{u_{i-1}}}{|\overrightarrow{u_{i-1}}|} \tag{8}$$

$$\overrightarrow{OQ_2'} = \overrightarrow{OP_i} + b\frac{\vec{u}_i}{|\vec{u}_i|} \tag{9}$$

where a and b are the offsets of $\overrightarrow{P_{i-1}P_i}$ and $\overrightarrow{P_iP_{i+1}}$, calculated by equations (6) or (7). The intersection point P_i' of $\overrightarrow{Q_1'P_i'}$ and $\overrightarrow{P_i'Q_2'}$ can be calculated as follows:

$$\begin{aligned}\overrightarrow{OP_i'} &= \overrightarrow{OQ_1'} + \lambda_1\overrightarrow{P_{i-1}P_i} \\ &= \overrightarrow{OQ_2'} + \lambda_2\overrightarrow{P_iP_{i+1}}\end{aligned} \tag{10}$$

where λ_1 and λ_2 are real numbers. Solving equation (10), the parameters of λ_1 and λ_2 are obtained as follows:

$$\lambda_1 = \frac{\left(\overrightarrow{Q_1'Q_2'} \times \overrightarrow{P_iP_{i+1}}\right)\cdot\left(\overrightarrow{P_{i-1}P_i} \times \overrightarrow{P_iP_{i+1}}\right)}{\left(\overrightarrow{P_{i-1}P_i} \times \overrightarrow{P_iP_{i+1}}\right)\cdot\left(\overrightarrow{P_{i-1}P_i} \times \overrightarrow{P_iP_{i+1}}\right)} \tag{11}$$

$$\lambda_2 = \frac{\left(\overrightarrow{Q_2'Q_1'} \times \overrightarrow{P_{i-1}P_i}\right)\cdot\left(\overrightarrow{P_iP_{i+1}} \times \overrightarrow{P_{i-1}P_i}\right)}{\left(\overrightarrow{P_iP_{i+1}} \times \overrightarrow{P_{i-1}P_i}\right)\cdot\left(\overrightarrow{P_iP_{i+1}} \times \overrightarrow{P_{i-1}P_i}\right)} \tag{12}$$

The offset point P_i' of p_i can be calculated by substituting λ_1 and λ_2 into equation (10). Using this method, the offset contours of the region cell A_m^j can be obtained. This method does not need to know whether the vertex of a cross-sectional contour is concave or not, and so the amount of calculation required is reduced.

关键是如何解决自相交的问题？

3.2.2 Elimination of self-intersections and loops

The generated offset contours of a region cell are likely to self-intersect and consist of a number of offset edges that do not contribute to the final offset contours. Therefore, they must be removed to ensure that the offset contours possess the features of a cross-sectional contour. The method presented by Koc and Lee (2002) is employed to remove the self-intersections and loops, but it is somewhat modified in that the offset vectors of the edges are used instead of the projected normals.

3.2.3 Offset part region generation

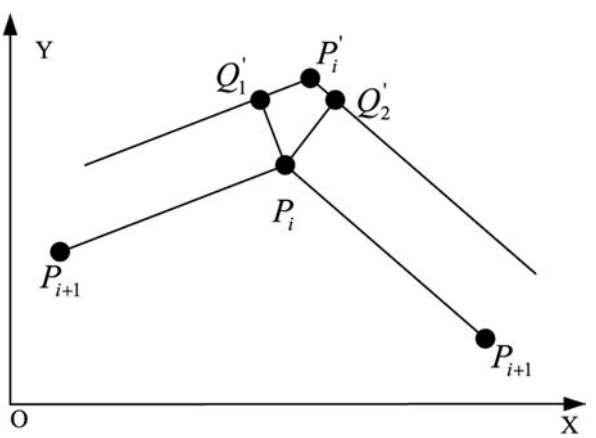
The offset contours of a given region cell A_m^j can be used to make corresponding polygons, after the self-intersections and loops have been removed. The anti-clockwise offset contours of A_m^j must be the external contours of the offset region cell $A_m^{j'}$ of A_m^j, denoted as $f^{j,m'} = \{f_1^{j,m'}, f_2^{j,m'}, \ldots, f_e^{j,m'}\}$, and the clockwise offset contours of A_m^j must be the internal contours of $A_m^{j'}$, denoted as $C^{j,m'} = \{C_1^{j,m'}, C_2^{j,m'}, \ldots, C_g^{j,m'}\}$.

Let $F^{j,m'} = \{F_1^{j,m'}, F_2^{j,m'}, \ldots, F_e^{j,m'}\}$ and $C^{j,m'} = \{C_1^{j,m'}, C_2^{j,m'}, \ldots, C_g^{j,m'}\}$ be the polygon sets for the polygons made by the contours in set $f^{j,m'}$ and $c^{j,m'}$, respectively. Then the offset region cell $A_m^{j'}$ can be calculated as follows:

$$A_m^{j'} = \bigcup_{t=1}^{e} F_t^{j,m'} - \bigcup_{t=1}^{g} C_t^{j,m'} \tag{13}$$

If $A_m^{j'} = \Phi$, it indicates that the offset region cell of A_m^j does not exist according to the given hollowing thickness (that is, in this case the cross-sectional contours on A_m^j were not offset in order to avoid

Figure 4 Offset contour point generation



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damaging the prototype). Once all the offset region cells on a given layer l_i are obtained, the offset part region PR'_i of PR_i on the layer l_i can be calculated as follows:

$$PR'_i = \bigcup_{j=1}^{\mathrm{Num}} \bigcup_{m=1}^{q} A'^{j}_{m} \tag{14}$$

However, in the cross-sectional contour offsetting operation, there is an exception when an edge of a cross-sectional contour is on the horizontal facet. In this case, solving equations (6) or (7) will cause a problem, because the offset of this edge is not in the horizontal plane.

'...In order to reduce the number of the horizontal facets and supports, more attention should be paid to the selection of the proper orientation in which the prototype is built...'

In this case, the cross-sectional contour offsetting operation on the layer where this edge lies will be skipped to the next layer, that is, the offset part region on this layer cannot be obtained by cross-sectional contour offsetting operation. Details of generating offset part regions on the layers inside an action region are discussed as follows.

3.3 Offset part regions generation inside action region

Generally, the layer thickness of a prototype is 0.1-0.2 mm in RP, and hence an STL model's cross-sectional contours on adjacent layers possess resemblances. In this work, the intersection operation is performed on the offset part regions made by all the offset contours on adjacent layers to obtain their identical areas. Given an action region $ar_i^u = (z_1, z_2)$ and $ar_i^u \in AR^u$, let the layer thickness be denoted as lt; let ar_i^u's in adjacent layers, whose Z-axis value is within $(z_2, z_2 + lt]$ and $[z_1 - lt, z_1)$ be denoted as l_{ui} and l_{di}, respectively; PR'_{ui} indicates the offset part region on the layer l_{ui} and PR'_{di} indicates the offset part region on the layer l_{di}. If l_{ui} or PR'_{ui} does not exist, let $PR'_{ui} = \Phi$; if l_{di} or PR'_{di} does not exist, let $PR'_{di} = \Phi$. Thus, the offset part regions on the layers inside the action region ar_i^u can be obtained by performing the intersection operation on PR'_{ui}, PR'_{di} and the offset part regions. These offset part regions are on the layers inside ar_i^u and are obtained by a cross-sectional contour offsetting operation. The details are described as follows.

(1) finding the offset part region PR'_{ui} and PR'_{di}, and

(2) if $PR'_{ui} = \Phi$ or $PR'_{di} = \Phi$, the layers inside the action region ar_i^u cannot have an offset part region.

Otherwise, the cross-sectional contour offsetting operation on the layers inside ar_i^u is used to obtain the offset part regions. Let the result of the intersection operation on these offset part regions be denoted as $PR_{ar'_i}$, then the offset part region PR'_i on each layer inside ar_i^u can be calculated as follows.

$$PR'_i = PR'_{ui} \bigcap PR'_{di} \bigcap PR_{ar'_i} \tag{15}$$

In order to reduce the number of the horizontal facets and supports, more attention should be paid to the selection of the proper orientation in which the prototype is built, because the offset part region derived from equation (15) may result in a hollowed STL model with an internal staircase effect.

Finally, after the orientations of the contours making the offset part region are reversed on each layer, these contours, together with the STL model's cross-sectional contours, constitute the cross-sectional contours of the hollowed STL model.

Algorithms for generating the cross-sectional contours of a hollowed STL model are described as follows:

```
Procedure hollowing operation()
  Begin
    Find horizontal facet
    Calculate action region using equation (1)
        or (2)
      Calculate action region AR^u using
        equation (5)
    For each layer l_i outside AR^u
      Call cross-sectional contour offset()
  End {for}
      Call action region offset()
  End

Procedure cross-sectional contour offset()
  Begin
    For each local MAX region F_j
    For each region cell A^j_m of F_j
    For each external and internal contour
      of A^j_m
      For every two adjacent contour edges
      Calculate offset value using equation
        (6) or (7)
      Calculate offset contour point using
        equation (10)
  End {for}
  Remove the self-intersections and loops
  End {for}
  Calculate A'^j_m of A^j_m using equation (13)
  End {for}
  End {for}
```

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Calculate PR'_i using equation (14)
Add the contours of PR'_i
End

Procedure action region offset()
Begin
For each ar_i^u
Find PR'_{ui} and PR'_{di}
If $PR'_{ui} \neq \Phi$ and $PR'_{di} \neq \Phi$ then
For each layer l_i inside ar_i^u
Call cross-sectional contour offset()
End {for}
Calculate PR'_i using equation (15)
For each layer inside ar_i^u
Add the contours of PR'_i
End {for}
End {if}
End {for}
End

These algorithms are used to construct the cross-sectional contours of the hollowed STL model for the RP process. Practical cases are discussed in Section 4.

4. Practical cases

Adopting VC++6.0, an offsetting system for RP has been developed according to the algorithms shown above. It is built into the data preparing system (DPS) 4.2 developed by the Institute of Advanced Manufacturing Technology (IAMT) in Xi'an Jiaotong University, China.
Two solid CAD models, one (Figure 5(a)) possessing surfaces nearly parallel to the slicing planes and the other (Figure 5(b)) having free-form surfaces were converted into the STL format using Unigraphics®, and then entered into DPS 4.2.

'...This hollowing operation has obtained satisfactory results in practical cases. The use of resin is reduced greatly; the building efficiency is enhanced and also the cost of production is reduced...'

After slicing the STL models and using the offsetting process, the offset contours can be checked in the contours editor. Figure 6 shows all the contours on layer 1665 of the artificial proximal femur. Figure 7 shows a series of contours from inside and outside an action region. The layers 1627-1726 have the same offset contours, which are the contours of the offset part

Figure 5 (a) 3D view of part I (a stamp); and (b) 3D view of part II (an artificial proximal femur)

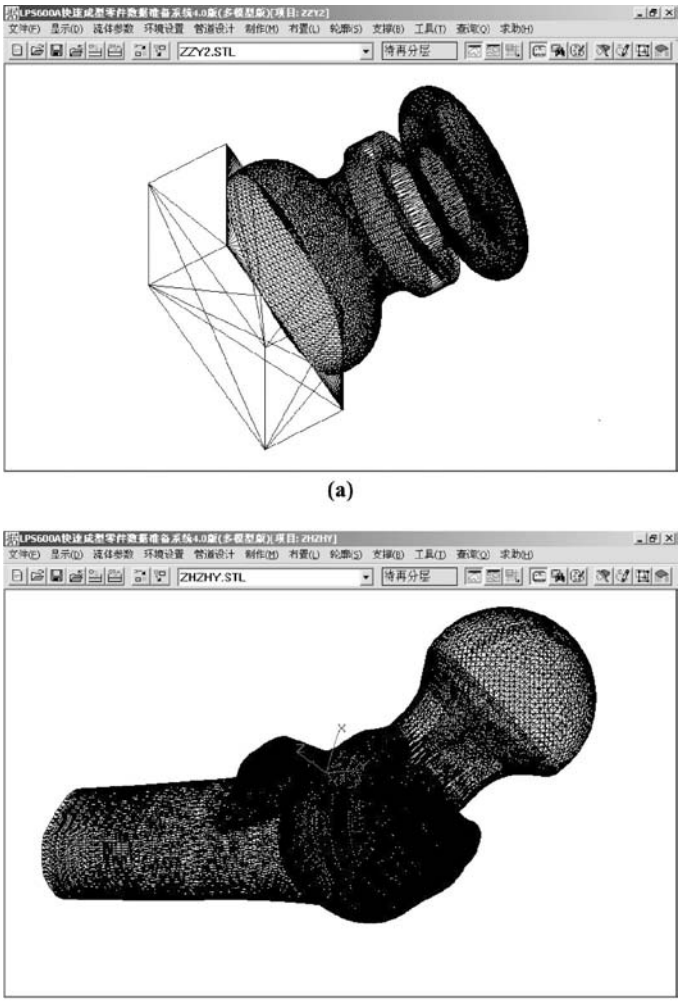
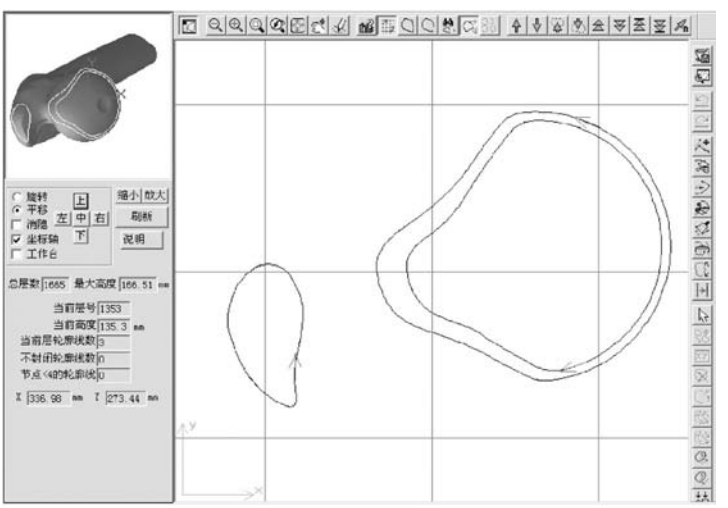


Figure 6 Cross-sectional contours of the artificial proximal femur



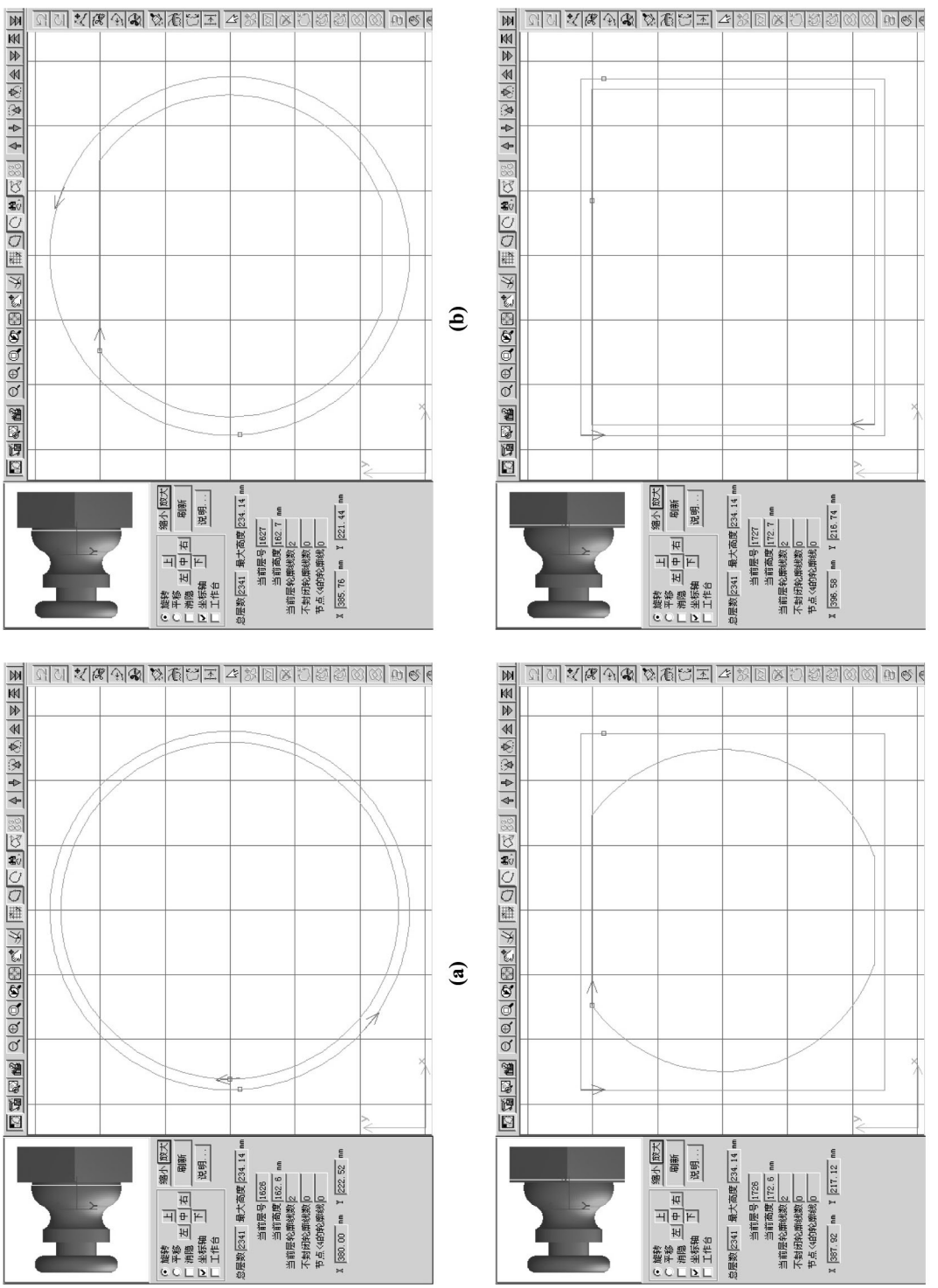
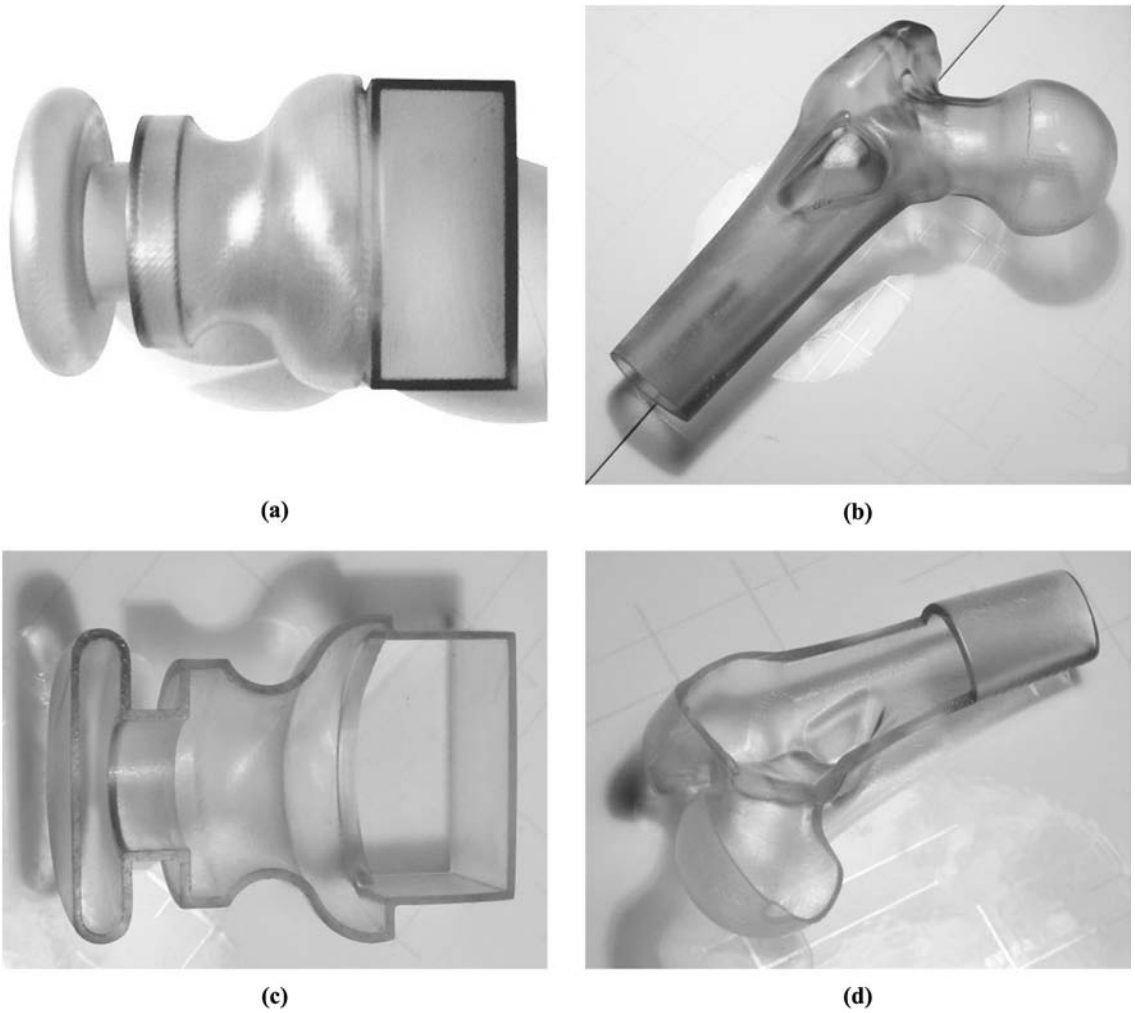


Figure 7 (a) Contours on the 1626th layer; (b) contours on the 1627th layer; (c) contours on the 1726th layer; and (d) contours on the 1727th layer

Figure 8 (a) Prototype of the stamp; (b) prototype of the artificial proximal femur; (c) vertical section of the stamp's prototype; and (d) vertical section of the artificial proximal femur's prototype



region derived from equation (15), because they are inside an action region.

After supports have been generated, the fileszzy2.lps and zhzhy.lps are output to build RP parts in the Laser RP Machine LPS600B. The prototypes and vertical sections taken through them are shown in Figure 8.

5. Conclusions

This paper has presented research on a new method of hollowing STL models for RP. Although this method cannot achieve a uniform wall thickness, it solves the problem of hollowing STL models with free-form surfaces. Considering the aforementioned applications for hollowed prototypes, this hollowing operation has obtained satisfactory results in practical cases. The use of resin is reduced greatly; the building efficiency is enhanced and also the cost of production is reduced.

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